

Review

Scales in single root water uptake models: a review, analysis and synthesis

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Summary

Scales in transport of water to roots are compared with the length and volume scales by using the concepts associated with the representative elementary volume (REV). The possibility of a mismatch between model scale and system scale when using a Darcy-Buckingham-based model to describe soil water transport to a single root is evaluated. In the absence of a mismatch, the replication requirements for evaluating the Darcy-Buckingham-based model near a single root are discussed by using a synthesis of the elementary scales involved, including those for soil, plant and roots, and of the measurement device. By using REV scales from lattice-Boltzmann simulations, the effective half-root mean distance and the available measurement techniques, the evaluation of Darcy-based single root uptake models is possible in roughly 50% of the combinations of soil- and root-system properties. On the basis of an assessment of the scale characterizing natural soil variability, the number of replicates required to assess the average root water uptake profile near a single root is large, and either requires miniaturization of the measurement methods for the hydraulic transport characteristics, or very homogeneous (artificial) growing media with little variability. Variability of water uptake per unit root length will increase the number of samples required.

Introduction

The use of single root models has a long tradition in studies related to both nutrient uptake (Baldwin *et al.*, 1973; Nye & Tinker, 1977; Barber, 1984; Hoffland *et al.*, 1990; Roose *et al.*, 2001), and water uptake by plant roots (Gardner, 1960; Cowan, 1965; Hillel *et al.*, 1975; Taylor & Klepper, 1975; Lafolie *et al.*, 1991; Personne *et al.*, 2003; de Jong van Lier *et al.*, 2006). In these models, the single root is defined as a linear sink with a radial flow pattern and homogeneous boundary conditions in the surrounding soil cylinder. At present the combination of models of root architecture and finite element numerical models of water flow allows studies of the fine structure of root water uptake by plants, explicitly and in detail, and over scales from single roots to a whole root system (Wu *et al.*, 2007; Javaux *et al.*, 2008; Pohlmeier *et al.*, 2008; Schneider *et al.*, 2010).

Experimentally, Dunham & Nye (1973) measured soil water extraction profiles near a root plane and Hainsworth & Aylmore (1986) and Crestana *et al.* (1985) measured soil water content

at high resolution in the immediate vicinity of an individual root. Development of techniques for high resolution measurements is ongoing and measurement resolution has increased further (MacFall *et al.*, 1991; Pohlmeier *et al.*, 2008; Segal *et al.*, 2008). The main experimental result is an expression of water content as a function of the distance from the root surface (extraction pattern or extraction cone) of the type shown in Figure 1, in one and two (radially-symmetrical) dimensions. The availability of sufficiently detailed measurements of soil water content profiles near a root would allow the evaluation of single root models and also that of spatial uptake patterns of a root architecture model.

In this approach to water uptake by plant roots, an interesting issue is whether the volume and the associated scale over which the pore-scale transport is averaged to yield results as described by the Darcy-Buckingham equation (the model scale) are compatible with the scale observed for transport towards a single root (the system scale). Issues of scale in transport towards the root were raised as early as 1969 by Nye in Klute & Peters (1969). Characteristic lengths in root water uptake are analysed by Raats (1990).

If the system scale is much smaller than that of the model (Darcy), it should be questioned whether transport towards a

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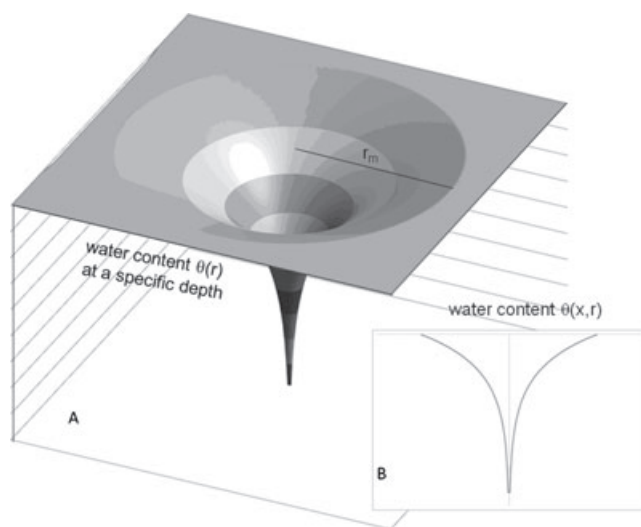


Figure 1 A schematic representation of the expected spatial distribution of water content around a root in two- (a) and one-dimensional (b) representations, assuming homogeneous properties around the root with radius r_m .

single root should be modelled at the pore scale with the Navier-Stokes equation. In this discussion, it could be argued that the Navier-Stokes approach should be used when modelling transport to a single root, given that the scales are generally very small, and that averaging Navier-Stokes over a volume of a porous medium should result in the Darcy equation (Vogel & Roth, 2003; Nordbotten *et al.*, 2007). In the present discussion, we take the position that models or the differential equations on which they are based are neither valid nor invalid but their performance when tested against system measurements may differ and their usefulness as a tool may change. Oreskes *et al.* (1994) and Kleindorfer *et al.* (1998) discuss these more philosophical aspects of model validity. We think that a critical analysis of scales involved in modelling water transport towards a single root using a Darcy-Buckingham approach is required.

An important concept in the discussion of the scales associated with the Darcy equation is the representative elementary volume (REV: Bear, 1972; Bachmat & Bear, 1986). It is defined as the smallest volume for which very small scale effects, caused by grain and pore size, do not affect macroscopic hydraulic properties and for which the continuum approach is still valid (Kutilek & Nielsen, 1994). Following Du & Ostoj-Starzewski (2006), we define an REV for which the continuum approach is valid as the Darcy-Buckingham elementary volume (V_D [L^3]). Given random resamples of this volume, the macroscopic flow (at the scale of the V_D) becomes proportional to imposed boundary conditions from the gradient in hydraulic head between two opposite boundaries of the V_D or an imposed constant flux boundary.

The REV concept is also used to define a volume that is statistically representative of the porous medium, and for which the variance of a property associated with this volume is locally minimal or locally not increasing. We will define this volume as the

statistical elementary volume (V_S [L^3]). The REV (as either V_S or V_D) is an interesting threshold for the study of soil water transport, whichever definition is used. One of the challenges in applying the Kutilek & Nielsen (1994) definition of the REV is that of 'not affecting macroscopic hydraulic properties'. A threshold of a 5% coefficient of variation in the variable of interest can be used (Pan *et al.*, 2001). Research in this area is continuing along the two approaches described above: either analysis of the statistical distribution of properties of the medium as a function of the volume considered (V_S), or analysis of the minimal averaging volume (V_D) for which estimates of variables used in a specific model first converge towards a stable value. This assumes that an ideal instrument is available that can measure hydraulic properties at any scale.

There is also a smallest scale at which soil hydraulic properties can be measured, labelled the instrument elementary volume (V_I , L^3). This volume is defined by the instrument that is actually used. For destructive sampling of soil water content (on a volume basis) the instrument elementary volume is well defined; for other instruments such as time domain reflectrometry (TDR), the measurement volume was analysed by Ferré *et al.* (1998). An analysis of effects of volume of tensiometer cups is presented by Hendrickx *et al.* (1994). The sphere of influence of neutron probes is inversely proportional to the cubic root of volumetric water content (IAEA, 1970). Luster *et al.* (2009) review modern measurement techniques in the rhizosphere, their applicability for laboratory and field measurements and their compatibility with modelling scales in root studies.

The single root uptake models are defined in terms of radial flow towards a long root. Therefore, the volume of the cylinder surrounding a single root (V_r [L^3]) is a relevant characteristic, determined by the effective half-root distance r_m [L] and the rooting depth Z [L]. The radial flow is additionally defined by the condition at the boundary between the soil and the root. In single root uptake models this boundary condition is uniform over the length Z . For a more general discussion it is useful to introduce a representative elementary root length z_b [L], which can be defined as the length over which the variance of the root-soil boundary condition is locally minimal or non-increasing. This length is a natural choice for root architecture discretization. In addition to the spatial scales (z_b and Z), the temporal scale should be considered. The measurement should not take longer than the process itself; characteristic time constants are the drying-out time t_d [t] and the measurement period t_i [t].

In view of this, potential scale issues in modeling the water uptake by a single root should be discussed in terms of the relative sizes of V_S , V_D , V_I and V_r , characterized by length scales $L_S = V_S^{1/3}$, $L_D = V_D^{1/3}$, and $L_I = V_I^{1/3}$ for the elementary volumes, and r_m , z_b and Z for V_r . In terms of temporal resolution, the period in which a cylinder dries out should be compared with the time required to measure a water content profile near the root. An illustration of some of the required spatial scales is given in Figure 2.

Four questions are of particular interest. Firstly, given the concept of the Darcy-Buckingham elementary volume V_D , are

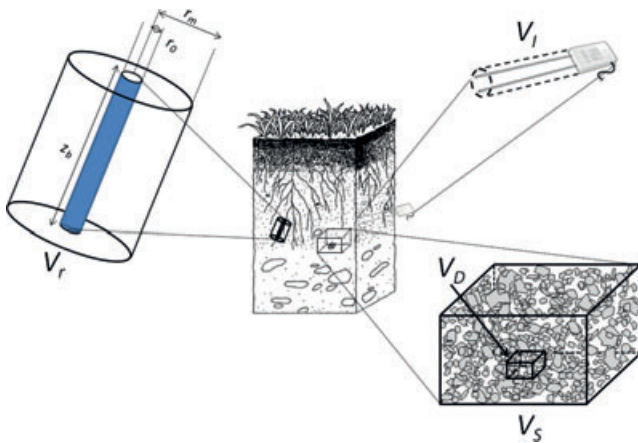


Figure 2 The soil elementary volumes, statistical elementary volume (V_s , 100 cm³), the Darcy-Buckingham elementary volume (V_D , 0.1 cm³), the instrument elementary volume (V_I , smallest volume 10⁻⁶ cm³) and the root characteristics (r_0 , 0.05 cm; r_m , 1 cm). The root elementary volume V_r is either based on the length z_b (the length over which root transport properties are least variable) or on the rooting depth Z . The example given here is based on the length z_b and is of the order 10 cm³; for the rooting system the volume is at least an order of magnitude larger. Values given are indicative only.

the volume and the characteristic scale associated with a single root large enough to evaluate experimentally Darcy-Buckingham-derived simulation results at this single root scale? Secondly, given the representative elementary root length z_b , does this length, over which the boundary condition is least variable, match or exceed the length scale associated with the Darcy-Buckingham elementary volume? If so, and thirdly, if measured soil water contents from the root surface to the outer bound of the single root volume are available, what is the number of replicates of these soil water content profiles required to calculate an average soil water content profile that would be simulated using soil hydraulic properties measured at scale V_s ? Finally, given that the spatial scales allow measurement, is the measurement period small relative to the rate of change in the particular variable being measured such as water content?

To discuss these four issues, we analyse the relationships between root-system parameters, the volumes V_s and V_D , and instrumental scale V_I .

Overview of scales in the analysis of uptake by a single root

Instrument scale, scale of the root system, statistical scale and Darcy scale

To characterize root water uptake patterns near a single root, the relevant length scale is that of the effective half root distance r_m , which is related to the root length density λ [L⁻²] as

$$r_m = (\pi\lambda)^{-\frac{1}{2}}. \quad (1)$$

The assumption is that a root system consists of a large number of identical parallel line sinks (Gardner, 1960; Cowan, 1965). The effective half distance between roots (r_m) and the rooting depth Z define the volume of soil explored by a single root V_r , as

$$\begin{aligned} V_r &= \pi r_m^2 Z \\ \text{or} \\ V_r &= \frac{Z}{\lambda}. \end{aligned} \quad (2)$$

The boundary condition (constant head or constant flux) at the soil system bound is often assumed to act over the entire rooting depth Z . Spatial and temporal variability within the root system strongly suggest that it is more realistic to assume that this boundary condition acts over a representative elementary root length z_b . As the root imposes a boundary condition on the surrounding soil, the length z_b should be larger than the length scale associated with the Darcy volume for the flow to become directly proportional to the imposed gradient after random re-sampling. Thus,

$$z_b > V_D^{1/3}. \quad (3)$$

Assuming one-dimensional radial flow, the Darcian hydraulic state variables have a non-linear dependency on the radial distance from the root centre r [L] near the root (Gardner, 1960; Cowan, 1965). The expected relationship between water content and spatial coordinates is illustrated in Figure 1 and will be referred to as the water extraction cone. As three points exactly define a quadratic function, the simplest approximation of non-linear dependency, at least three observations of these state variables are required at different values of r . Consequently, r_m needs to be at least $3V_D^{1/3}$ or $3V_I^{1/3}$, whichever is largest. Therefore, defining R [–] as

$$R = \frac{r_m}{\max(V_D^{1/3}, V_I^{1/3})} = \frac{r_m}{\max(L_D, L_I)}, \quad (4)$$

A requirement that $R \geq 3$ has to be met in order to evaluate experimentally the hydraulic state variable profile from the root surface to the outer boundary of the single root volume, predicted by using a Darcy-Buckingham-based solution. On the basis of the definitions introduced above, V_D is the minimum size at which Darcy-Buckingham-type patterns become observable, and the volume V_s is the minimum size over which these patterns should be observed (measured) to capture a value that is representative of the application considered. Often V_s is implicitly defined by the volume used to determine experimentally the variable of interest such as hydraulic conductivity.

The variances of the properties in V_s and V_D can be related, according to an approach based on geostatistical analysis by Zhang *et al.* (1990), to fractal dimensions (Bartoli *et al.*, 2005) and used as an empirical equation relating yield variance to plot area in early experimental crop yield design

(Fairfield Smith, 1938). In this approach variance changes with volume as

$$\frac{\sigma_S^2}{\sigma_D^2} = \left(\frac{V_D}{V_S} \right)^b, \quad (5)$$

where V_S is the statistical elementary volume, V_D the Darcy elementary volume, σ_D^2 the variance of the variable of interest in the V_D and σ_S^2 the variance associated with the property in V_S . The parameter b reflects the effects of spatial heterogeneity: if b is positive, variance decreases with sampling volume. In terms of a (re)sampling requirement at scale V_D , Equation (5) can be rewritten as

$$n = \left(\frac{S_D}{S_S} \right)^{\frac{2}{b}}, \quad (6)$$

where n is the number of samples, S_D is the standard deviation of the soil physical variable associated with the volume V_D and S_S is the standard deviation of the same physical variable associated with a volume V_S . For $b = 1$, standard errors decrease as a function of sampling volume as if the data consist of independent contributions of volumes V_D within the volume V : for $b = 0$, the variance is not a function of volume. If $b > 1$, variance decreases strongly with volume, suggesting that a cause of variance disappears, as in soils with macropores. Given a b -value, this approach can be put to two uses: the number of replicates required to upscale from a sample at scale V_D to a larger volume can be calculated from a specific relative standard deviation (Equation (6)), and the expected reduction in variance when upscaling to a specific volume can be calculated with Equation (5).

Variability of the boundary condition along the root length

The simplest root water uptake model reported in literature assumes a binary spatial uptake pattern along a single root, with a maximum in the first 8–12 cm, which then decreases to a constant but much smaller value along the remaining length (Kolek & Kozinka, 1992; Segal *et al.*, 2008). In most current root-system architecture modelling, however, roots are differentiated into segments. For these segments, properties are assumed to be homogeneous, suggesting that boundary conditions along them are also homogeneous. Somma *et al.* (1998), for instance, assume that water uptake depends on root age and root branching order, and that uptake along a specific root segment is uniform.

This discretization of continuous properties is an approximation because physiological properties along a root vary continuously and radial and axial conductances change over time and location, as discussed by Steudle & Peterson (1998). Changing conductances were used by Doussan *et al.* (1998), who mathematically fitted time and spatially dependent patterns of conductances to optimize the behaviour of their model with respect to experimental data. Given these system properties there are three possibilities. The first is to analyse the length scales used in the discretization of current architectural modelling, and assume that the use of this

approach is justified. The second is to analyse the length of specific root types, such as the apical unbranched root segment length (for which the maximum uptake rate per unit length is expected) or lateral root lengths. Thirdly, root elongation rates could be analysed. When root lengths or elongation rates are analysed, it has to be assumed that the root types are homogeneous in their properties. The elongation rates ($L T^{-1}$) allow for root calculating, which, after time period (T), the root segment length exceeds the Darcy scale (L). In accordance with the definitions used above, we will refer to a root segment length with homogeneous conductances as the representative elementary root length.

Temporal resolution: process temporal scale and instrument temporal scale

The amount of water stored in the soil cylinder surrounding the root [L^3] divided by the potential uptake per root ($L^3 T^{-1}$), calculated from potential transpiration and the total number of roots, yields a minimum time scale (T) for uptake of water, which is relevant only if plants do not modify their uptake pattern when experiencing water stress. If plants do adapt to stress and reduce their uptake rate after some limiting water content has been reached, the period to zero flux (wilting) will increase. Metselaar & de Jong van Lier (2007) define the initial rate of the decrease after soil water content has started to limit transpiration as a characteristic time (t_d) that is dependent on soil physical properties, rooting depth and potential transpiration. An interesting simple case is that of an exponential diffusivity for which t_d is simply rooting depth Z divided by potential transpiration (T_p). If the scan time of an instrument is t_i , then t_i should be much smaller than t_d . If the instrument measurement error in water content is $\Delta\theta_i$, then $\Delta\theta_i t_d$ is the minimum time between measurements required for the instrument to resolve two subsequent water content measurements.

Literature review of spatial and temporal scales

The effective half mean root distance

Values characterizing the root systems of crops were presented by de Willigen & van Noordwijk (1987). Their data (excluding pot experiments) allow us to calculate an overall r_m of 7 mm with a variability between sources expressed as standard error of the mean of 0.7 mm ($n = 95$). The average rooting depth observed in the experiments is 84 cm (with a standard error of the mean of 8 cm; $n = 36$). Averaging the individual single root volumes, the effective volume explored by a single root is of the order of $180 \pm 60 \text{ cm}^3$.

The representative elementary root length

For maize, Pellerin & Pagès (1994) present a distribution of the apical unbranched axial root lengths, which suggests a minimum length of 15 mm, and a mode of 45 mm, depending on the elongation rate of the parent root. Their data also allow us to

estimate the initial axial root elongation rate for a number of maize elementary vegetative functional units (phytomers) as 3 mm per degree-day with a standard error of the mean of 1 mm ($n = 6$; all roots taken into account). The degree-day base temperature is 6°C. For lateral maize roots (Pagès & Pellerin, 1994), the length distribution was 25 mm (mean) and the median was 16 mm. Elongation rates were not estimated. In the model described by Somma *et al.* (1998), the three orders of roots were characterized by elongation rates of 24, 4.8 and 1.9 mm day⁻¹.

The Darcy-Buckingham elementary volume

The results available to define the minimum size associated with the Darcy volume (V_D) are at present based on the analyses of detailed pore scale lattice-Boltzmann models and the estimation of averaging volumes above which estimates of dynamic hydraulic variables such as conductivity become stable. In these cases, the volumes are defined as dimensionless quantities with a characteristic scaled length L_D^* , expressed in terms relative to the particle diameter (Pan *et al.*, 2001), in terms of the correlation length (Zhang *et al.*, 2005), or the repetition length (Pilotti *et al.*, 2002). An overview is given in Table 1. On the basis of these results, the length scale of the V_D in soils of a single grain-size ranges between 0.2 and 3 mm (2–5 periodic lengths, or 10–16 particle diameters). We are not aware of studies that analyse the V_D in structured fine-textured soils.

The statistical elementary volume V_S

In most papers that analyse REV in terms of a statistically representative sample size, the aspect examined is that of the existence of a (local) minimum, an inflection point or an asymptotic value in the relationship between variance and sample volume. This analysis leads to a diverging range of values, varying from 3 mm³ as established for Castlegate sandstone (Fredrich *et al.*, 2006) to 400–2100 cm³ for a stony soil (Buchter *et al.*, 1994). Kutilek & Nielsen (1994) suggest an REV of 100 cm³

(equivalent to an edge length of L which is of the order of 5 cm) for single grain-size structures, and a value one or more orders of magnitude greater for strongly aggregated soils.

As summarized in Table 1, the reported characteristic edge lengths L_S for V_S are in the range of 0.2–260 mm, with the larger values for variables such as stoniness and bulk density. The smaller values are based on studies of simulated spherical packing. In these cases the dimensionless scaled characteristic length L_S^* (edge length relative to particle diameter) is sometimes used. The mean volume V_S based on these data is 40 cm³ but the median volume is only 100 mm³.

Experimental resolution

The instrument length scale (V_I)^{1/3} with which water contents near the root surface can be determined has decreased over the years. One of the first measurements of gradients near roots was presented by Dunham & Nye (1973), who measured water contents with a minimum sample thickness of 2.3 mm. Hainsworth & Aylmore (1986) used gamma attenuation and achieved a sampling distance of 1.5 mm. More recently, Hamza *et al.* (2001), using X-ray attenuation, achieved a resolution of 0.1 mm. A resolution of 0.1 mm was also achieved in an MRI scan (MacFall *et al.*, 1991). There is a trade-off between spatial resolution and errors in moisture content. An error analysis for neutron tomography by Tumlinson *et al.* (2008) led them to conclude that given a spatial resolution of 80 µm, an average length of 0.5 mm results in acceptable moisture content errors (0.01 m³ m⁻³). At present, the minimal experimental resolution is 0.1 mm, with an associated measurement volume (V_I) of 0.001 mm³. This spatial resolution is often achieved at the cost of temporal resolution, but both are also determined by sample size. Perret *et al.* (2000), in a tracer experiment, quote a temporal resolution of 2 ms with a spatial resolution of 6 mm (single photon emission computed tomography, SPECT), and a temporal resolution of 2 s with a spatial resolution of 0.2 mm (computerized axial tomography, CAT). Reduction of this resolution may be

Table 1 Synthesis of data on the different scales relevant in transport of water towards a single root

	Mean	Standard deviation	Minimum	Lower quartile	Median	Upper quartile	Maximum	Number of observations	Source ^a
r_m / mm	7.0	6.8	1.1	3.4	5.0	8.0	39.9	95	1
L_D / mm	1.6	1.2	0.2	0.8	2.0	2.1	3.1	5	2
L_D^* (ratio)	7.5	5.9	1.5	—	5	—	16.2	5	2
$R = r_m / L_D$	11.5	22.4	0.4	1.9	3.8	10.6	221.6	475	—
L_S / mm	35.0	59.3	0.3	2.7	3.8	32.5	262.1	36	3
L_S^* (ratio)	7.0	4.3	2.5	—	6.6	—	12.6	4	4
$(V_D/V_S)^b$; $b = 1$	18	156	3.2×10^{-10}	2.4×10^{-5}	1.6×10^{-3}	0.3	1940	180	—

^aSource: numbers between square brackets denote respective number of observations. 1. de Willigen & van Noordwijk (1987) [95]. 2. Zhang *et al.* (2005) [1]; Pilotti *et al.* (2002) [2]; Pan *et al.* (2001) [2]. 3. Clausnitzer & Hopmans (1999) [1]; Vogel *et al.* (2002) [1]; Baveye *et al.* (2002) [4]; Buchter *et al.* (1994) [4]; Al-Raoush & Willson (2005) [6]; Bartoli *et al.* (2005) [2]; Mayer & Miller (1992) [6]; Razavi *et al.* (2007) and Razavi (2006) [3]; Fredrich *et al.* (2006) [1]; Martinez *et al.* (2007) [8]. 4. Clausnitzer & Hopmans (1999) [1]; Razavi *et al.* (2007) and Razavi (2006) [3]. Numbers in italics indicate the total number of combinations assuming no correlation between variables used in the ratio.

expected in the future due to the improvement and miniaturizing of electronics and possibly new techniques such as the use of micro-biosensors (genetically altered bacteria that can be used to provide information at the microbial scale) (Cardon & Gage, 2006).

Variance as a function of sampling volume

The variance of soil properties in terms of Equation (5) has been assessed as a function of either area or volume. Anderson & Bouma (1973) analysed the value, and the variance, of saturated conductivity (K_s) as a function of volume (ranging from 150 to 1700 cm³) and presented results yielding a b -value of 1.12 ± 0.24 ($n = 4$), equal to one for all practical purposes. Results presented by Clothier & White (1981), however, for a number of soil physical variables and two sample volumes (435 and 5301 cm³) suggest that variance may increase as a function of sample volume. The analysis of soil nitrate (Starr *et al.*, 1992) and denitrification activity (Parkin *et al.*, 1987) also results in much smaller values of b , in the range of 0.1–0.2 for denitrification and –0.1 (in other words, increasing) to 0.4 for nitrate. A study of the variability of soil water content as a function of sample size yields a value of 0.3–0.5 (Hawley *et al.*, 1982). Recent studies of water content profiles near a root suggest that on these small scales soil physical properties may change to such an extent that the profiles are inconsistent with the expected direction of water flow (Nakanishi *et al.*, 2005; Carminati *et al.*, 2010).

At present, the available data for soil physical properties suggest variance as a function of sampling volume to be both variable and very dependent on the property of interest. We know of no results available that define a b -value spanning the range between V_D and V_S for soil physical variables such as soil water content.

Temporal scales

Macroscopic soil water balance studies are typically performed at the 1-day scale, whereas microscopic (single-root) modelling involves time scales that are often shorter. At the root surface, de Jong van Lier *et al.* (2006) simulated differences in pressure head of >100 m over a 12-hour period, corresponding to a difference in water content of 0.20 m³ m^{–3}, or a $d\theta/dt$ of 4.6×10^{-6} m³ m^{–3} s^{–1}. If a typical instrument error in the order of 0.01 m³ m^{–3} for soil water content sensors is assumed (Rosenbaum *et al.*, 2010), the variation of water content over approximately 30 minutes would be detected reliably by the instrument. Most sensors for soil water content make their measurements in a matter of seconds and no limitations to the measurement capacities of these sensors should be expected to arise from temporal resolution. This may be different in 3-D tomography measurements, which may take more time. In these cases, length of time of measurement may become a critical aspect.

Synthesis: combining spatial scales

The synthesis of data for root and soil scales is presented in Table 1. Important derived auxiliary components are the ratio R

(Equation (4)) and the ratio $(V_D/V_S)^b$ (Equation (5)). We assume that measurement resolution is not restricting the analysis, as V_I is much smaller than V_D . In that case, R equals r_m/L_D . The ratios r_m/L_D and $(V_D/V_S)^b$ were calculated for all possible combinations of individual observations; note that the mean of these ratios is not equal to the ratio of the means. The ratio r_m/L_D is the number of independent measurements possible along the flow path toward the root as determined by the experimental system. On the basis of the data presented in Table 1, the criterion that the root length scale z_b should be larger than the edge length of the Darcy-Buckingham volume L_D is generally met, both for the root elongation rates and the established root length (lateral and apical) for maize. Results suggest that the time steps for growth-rate calculations should not be less than one day, as minimum daily elongation rates are in the same order as L_D . In terms of spatial soil scales, the ratio $(V_D/V_S)^b$ is calculated, again for an assumed value of $b = 1$, which yields the expected reduction in variance when upscaling from the Darcy elementary volume to the statistical elementary volume.

Assuming that instrument resolution is not restricting the analysis, the median of the ratio r_m/L_D is 3.8, which means that the scales are such that four or more spatially independent observations of the non-linear radial water content profiles towards a root are measurable in the Darcy-Buckingham framework in at least half of the investigated experiments shown in Table 1. The median of the ratio $(V_D/V_S)^b$ (for $b = 1$) is 1.6×10^{-3} and thus in half of the cases variance of an (averaged) soil property is reduced to 1/600 or less of the variance associated with the smallest sample volume.

Discussion and conclusions

We have presented an analysis of elementary volumes for use in single-root water uptake models. For upscaling to the root-system scale, which is beyond the scope of this paper, these uptake models need to be coupled to a description of root-system structure. This can be achieved by calculating the volume exploited by each single root from the mean distance between roots as a function of the spatial distribution of root-length density (de Jong van Lier *et al.*, 2006). According to de Willigen & van Noordwijk (1987), root length densities generally range between 0.5 and 5 cm cm^{–3}, corresponding to a mean radius of exploitation of 2.5–8.0 mm. In reality, however, a large variation of exploitation radii will exist within one root system. These can be characterized by fractal analysis (Tatsumi *et al.*, 1989) or by root-growth modelling (Diggle, 1988; Doussan *et al.*, 1998) and then coupled to a single-root approach. In our analysis, we have also excluded issues of spatial and temporal heterogeneity in the root architecture and in the root uptake characteristics.

Given the above information, within the limits of this study, and not considering requirements for replication, the probability is 50% that, in a given experiment (defined by the edge length of the Darcy-Buckingham elementary volume L_D , and the effective half-mean distance between roots), at least four measurements

describing the moisture content towards a root can be made. Thus in half of the soil-plant combinations considered, the conditions to evaluate a Darcy-Buckingham approach to describe water transport to a single root are met. The edge length L_D is a very uncertain quantity, and at present can only be derived on the basis of lattice Boltzmann model simulations, which approximate the behaviour of the Navier-Stokes equation. Using the minimum root elongation rates from Somma *et al.* (1998), and comparing the resulting root scale (the root length z_b) with soil scales (edge length L_D) the minimum time-step when modelling root growth is one day, given that the root volume added would then correspond to the Darcy elementary volume. For the fastest growing roots this may be a factor of ten smaller.

If soil properties are not correlated, the data suggest that the water content measured in volumes required to sample water content near a root is 600 times as variable as the soil water content in the volumes normally sampled and assumed to be representative of the bulk soil (including roots). Presence of spatial correlation in soil properties will raise the variability, increasing this factor. This suggests that the experimental system, in terms of the number of independent spatial replicates, will be crucial in testing those models that predict water content profiles near roots, especially if hydraulic conductivity or retention curves are measured in volumes that are much larger than those defining the process scale. Unless water retention and conductivity properties can be determined for very small samples, or alternatively the growing medium is very homogeneous, the number of replicates required is very large (in the order of 600).

The issue raised by the observations that soil water content can also increase towards the root is that measuring soil water potentials would define unambiguously the water-flow direction, which makes miniaturization of tensiometers an essential requirement in understanding flow patterns near roots. The discussion in terms of scales should then refocus on the matric-head characteristics of the soil.

An alternative to validate a microscale model at the scale at which it has been formulated, is to test an upscaled microscopic model for an entire rooting system. A natural upscaling method is based on root architecture, which, however, has its own parameterization and replication problems. The use of complex root architecture models as a basis for deriving and testing macroscopic root-water uptake models with fewer parameters may be a way to solve both parameterization and scale issues for water uptake in agricultural practice.

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